Anna Cobb

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Research Interests

Technoeconomic modeling, environmental impact assessment, and policy analysis of transportation and energy technology usage and adoption. Current topics of interest include responsible management of lithium-ion batteries at end of life and supply and demand of critical minerals streams over time.

Education

Carnegie Mellon University (Pittsburgh, PA) **2022 – August 2026 (expected)**

- 4th year Ph.D. Candidate in Engineering and Public Policy
- Dissertation: "Social and Environmental Sustainability in Transportation: Analyses of Ride-Hailing Services and the Circularity of Electric-Vehicle Batteries"
- Advisor: Jeremy Michalek, PhD
- Committee Members: Jeremy Michalek (Chair), Jennifer Dunn, Alissa Kendall, Jay Whitacre
- National Science Foundation Graduate Research Fellow (NSF GRFP)

Georgia Institute of Technology (Atlanta, GA) **2018 – 2022**

- Bachelor of Science in Mechanical Engineering, minor in Energy Systems

Research Experience

Carnegie Mellon University (Pittsburgh, PA), *Graduate Research Assistant* **Since August 2022**

- Paper 1: Developed an agent-based model in Julia to simulate Uber and Lyft trips in for analysis of the impact of discriminatory driver behavior and matching algorithm parameters on wait times
- Paper 2: Created a process-based cost model of a battery repurposing facility, which was used along with empirical data and simulations of battery degradation to compare the economics of repurposing and recycling electric vehicle batteries across a wide variety of conditions
- Paper 3: Using the EverBatt recycling cost and emissions model to examine the net benefits of lithium-ion battery recycling across cathode chemistries, evolving feedstock, and material prices

TerraPower (Seattle, WA), *Hydrogen Generation Development Intern* **Summer 2022**

I conducted preliminary investigations of integrating hydrogen production technology with TerraPower's Natrium nuclear reactor design by writing a literature review and developing a systems-level technoeconomic model to evaluate different hydrogen production methods.

Ben T. Zinn Combustion Lab (Atlanta, GA), *Undergraduate Research Assistant* **2020 - 2022**

In collaboration with a master's student, I worked on the development of a probe for monitoring gas turbine engine health. I focused on the creation of post-processing code in MATLAB for identifying and characterizing thermoacoustic instabilities from the combustion rig test data.

Solar Turbines (Virtual), *Research Fellow* **Summer 2021**

Created and calibrate a geometric flow model of a test combustion rig. Model simulation results were used to help identify combustion instabilities that can occur during engine operation.

Skills

Programming Languages/Software

R, Python, Julia, MATLAB, GitHub, use of supercomputing resources

Models

EverBatt, BLAST-Lite, GREET R&D

Relevant Graduate Coursework

Engineering Optimization, Introduction to Machine Learning, Electrochemical Decarbonization Technologies, Multi Criteria Decision Making, Data Science for Technology and Policy

Academic Projects

USAEE Virtual Case Competition (Carnegie Mellon), *Group Member* **August 2025**
Month-long competition focused on strategies for utilities to handle load growth from data centers.
- Ran Monte Carlo simulations to examine how the composition of data centers requesting connection affects magnitude of uncertainty over time.
- Team Placement: 1^{st}

NASA Blue Skies Competition (Carnegie Mellon), *Group Member* **Academic Year 2022-23**
Year-long competition focused on developing and analyzing supply chain for an alternative aviation fuel.
- Assessed hydrogen supply chain technology readiness for use in commercial aviation in terms of cost, technology readiness levels, and emissions
- Supported development of multi-objective optimization model of hydrogen supply chain technologies including production, long-distance transport, and last-mile transport
- One of eight teams selected to compete in final competition at NASA headquarters in June 2023: awarded "Best Oral Presentation"

Georgia Tech EcoCAR (Georgia Tech), *PCM Sub-Team Lead* **2021 - 2022**
Four-year collegiate competition focused on converting a gas-powered Chevrolet Blazer to a hybrid electric vehicle through software and hardware reconfiguration, testing, and installation.
- MIL (Model-in-Loop), HIL (Hardware-in-Loop), and VIL (Vehicle-in-Loop) software testing
- Automated and improved robustness of existing MIL testing procedures
- Trained new team members, developed high-level team goals, delegated work to team members, and collaborated with leads at other universities
- Individual Placement (Propulsion, Controls, and Modeling (PCM) presentation): 1^{st}
- Team Placement: 1^{st}

Teaching Experience

Carnegie Mellon (Pittsburgh, PA), *Teaching Assistant* **Fall 2025**
Introduction to engineering and public policy – responsibilities: grading, hold weekly recitations

Carnegie Mellon (Pittsburgh, PA), *Teaching Assistant* **Fall 2024**
Engineering Optimization – responsibilities: conversion of course content from MATLAB to Python, assisting students with troubleshooting coding problems

Carnegie Mellon (Pittsburgh, PA), *Peer Tutor* **Spring 2024**
Introduction to Machine Learning – responsibilities: increasing comprehension of course content

Georgia Tech (Atlanta, GA), *Teaching Assistant* **Spring & Fall 2020**
Numerical Methods in MATLAB – responsibilities: grading, office hours, & review sessions

Awards & Honors

Robert W. Dunlap Award **Spring 2024**
Awarded for the most outstanding solution submitted for the Part B Qualifying Examination of the Department of Engineering and Public Policy

Carnegie Institute of Technology Presidential Fellowship **Academic Year 2024-2025**

Dean's Fellowship **Academic Year 2022-23**

National Science Foundation Graduate Research Fellowship **Spring 2022**
Prestigious fellowship providing funding for tuition & stipend for 3 years of graduate education. Awarded for research proposal focused on hydrogen supply chain development and optimization

President's Undergraduate Research Award **Spring 2021**
Research proposal: "PMT Data Analysis for Flame Transfer Function of Test Combustion Rig"

Leslie U. and Ola Ryle Hammack Memorial Scholarship **Academic Year 2020-21**
Awarded to junior-year mechanical engineering student at Georgia Tech with highest GPA who is a Georgia resident

Zell Miller Scholarship **2018 – 2022**
Full tuition coverage awarded to residents of Georgia for maintaining a 3.3 GPA

Media, Commentary, & Other Contributions

Volta Foundation 2025 Annual Battery Report (Contributor): Contributed analysis to the Downstream Supply Chain section of the report, focusing on legislation, market trends, and environmental benefits of battery recycling

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Volunteer Work

Graduate Application Support Program, *Team Member* **Academic Year 2025-2026**

- Match Engineering and Public Policy (EPP) graduate applicants with current students to provide support with the application process (Q&A, feedback on application materials)

GROW+ (Pittsburgh, PA), *Co-Facilitator* **Academic Year 2023-2024**

- Facilitate social, academic, and professional event planning to support women and non-binary graduate students in the engineering and public policy department

Student Competition Center Outreach Committee (Atlanta, GA), *Chair* **Academic Year 2021-22**

- Plan and execute youth outreach events for members of Georgia Tech's competition teams

Little Einstein's Organization (Atlanta, GA), *Member* **2021 – 2022**

- Plan and complete STEM activities with children at underserved schools in the Atlanta area

GT Haiti Solar Initiative (Atlanta, GA), *Team Member* **2018 – 2021**

- Wrote grant application and presented to board; earned $3,500 in funding for solar sewing project
- Completed development of prototype system and shipped components to Haiti in Summer 2019
- Visited high schools during Spring 2021 to teach students about Haiti's electrical grid and introduce them to the fundamentals of circuit-building

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Publications [Google Scholar Link]

In Preparation

A. Cobb *et al.*, "Lithium Ion Battery Recycling Economics and Externalities: Is policy intervention required?" *In preparation*, April. 2026.

Under Review

A. Cobb and N. Khanna, "Rethinking electric vehicle battery design: Policy levers for incentivizing integration of circularity design principles," *Under Review at Humanities and Social Sciences Communications*, April 2026.

Published

A. Cobb *et al.*, "Electric-vehicle battery second-life and recycling pathways: How economics depend on chemistry, processing, and application," *Applied Energy*, vol. 414, p. 127809, Jul. 2026, doi: 10.1016/j.apenergy.2026.127809

M. Tsuchiya, A. Cobb, and P. Vaishnav, "Chicago Riders' Choice of Uber and Lyft over Transit Implies a Median Breakeven Value of Travel Time Equal to the Regional Hourly Wage of $30 per Hour," *Environ. Sci. Technol.*, vol. 59, no. 4, pp. 1921–1931, Feb. 2025, doi: 10.1021/acs.est.4c08808.

A. Cobb, A. Mohan, C. D. Harper, D. Nock, and J. Michalek, "Ride-hailing technology mitigates effects of driver racial discrimination, but effects of residential segregation persist," *Proceedings of the National Academy of Sciences*, vol. 121, no. 41, p. e2408936121, Oct. 2024, doi: 10.1073/pnas.2408936121.

A. Matthews *et al.*, "Experimental Development of On-Line Flame Transfer Function Measurements for Fielded Gas Turbines," Sept. 2021. doi: 10.1115/GT2021-59317.

Oral Presentations

International Battery Seminar & Exhibit, *Retirement Planning for Electric Vehicle Batteries: Why lithium iron phosphate batteries have greater second-life potential than nickel-cobalt batteries*, Orlando, FL, March 26, 2026

Scholar-to-Scholar Graduate Project Showcase, *Lithium-Ion Battery Recycling Economics and Externalities: Is Policy Intervention Warranted*?, Pittsburgh, PA, March 20, 2026

Monthly Julia Dynamics Meeting, *Ride-hailing mitigates the effects of racial discrimination, but effects of residential segregation persist*, Virtual, February 13, 2026

Battery Recycling & Sustainability Summit, *Retirement Planning for Electric Vehicle Batteries: Why lithium iron phosphate batteries have greater second-life potential than nickel-cobalt batteries*, Bethesda, MD, November 13, 2025

248th Electrochemical Society (ECS) Conference, *Retirement Planning for Electric Vehicle Batteries: Why lithium iron phosphate batteries have greater second-life potential than nickel-cobalt batteries*, Chicago, IL, October 13, 2025

Scott Institute Student Seminar Series, *Retirement Planning for Electric Vehicle Batteries: Why lithium iron phosphate batteries have greater second-life potential than nickel-cobalt batteries*, Pittsburgh, PA, September 23, 2025

Volta Foundation Battery Forum, *Why lithium iron phosphate batteries have greater second-life potential than nickel-cobalt batteries*, Virtual, September 18, 2025 [available on YouTube]

Carnegie Mellon GROW+ Student Workshop, *Agent-Based Modeling for Dummies*, Pittsburgh, PA, March 12, 2024

Vehicle Electrification Group Meeting, *Using the Bridges 2 Supercomputer: Observations, Information, and How You Can Waste Less Time than Me*, Pittsburgh, PA, November 13, 2023

Poster Presentations

Steinbrenner Sustainability Symposium, *A Technoeconomic Analysis of Repurposing EV Batteries,* Pittsburgh, PA, March 28, 2025